Monthly Intelligence Report

Edition 001 — The Śląskie–Mazowieckie Industrial Corridor

Where coal transition is straining grid resilience. Inaugural edition · April 2026 · SSI v4.0.2

CONTENTS

SECTION A

Executive Summary

SUBSTATIONS SCORED

2,248

Major + distribution across PSE (Polskie Sieci Elektroenergetyczne)

GRID LINES MAPPED

~2,847

~52,380 km transmission & distribution

MC SIMULATIONS

~22.5 M

10,000 per substation

PUBLIC DATA SOURCES

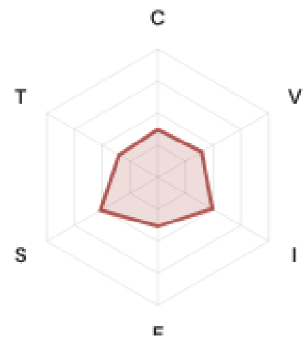
30+

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions. Its engine processes 254,700+ source data points per run, executes ~22.5 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 3.02 billion arithmetic operations — producing 717,595 output data points with full uncertainty quantification across 2,248 substations and ~2,847 grid lines spanning ~52,380 km.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30+ verified public sources (URE, PSE, PGNiG, IMGW-PIB, GUS, PERN, PAIiIZ), making the methodology fully auditable and reproducible. Initially covering Poland's PSE-managed transmission and five major distribution system operators, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus



Stacja elektroenergetyczna 110kV „Koksownia Zdzieszowice"

Opolskie, Opolskie · 110 kV

0.507	High	0.167	70th
R_FINAL	CLASSIFICATION	CI WIDTH	FLEET PERCENTILE

COMPONENTS

- **C** Continuity: 0.373 / 0.30 (124%)
- **I** Infrastructure: 0.493 / 0.25 (197%)
- **S** Saturation: 0.517 / 0.20 (259%)
- **V** Voltage: 0.397 / 0.10 (397%)
- **E** Economic: 0.384 / 0.10 (384%)
- **T** Transition: 0.349 / 0.05 (698%)

MODIFIERS

- **R3 Consequence:** 1.088 ↑ amplifies
- **R4 Graph Criticality:** 1.007 → neutral
- **R6a Restoration:** 0.982 ↓ dampens
- **R6b Seismic:** 1.061 ↑ amplifies
- **R7 Digital Readiness:** 1.029 ↑ amplifies

This month's spotlight — automatically selected for June 2026 — is **Stacja elektroenergetyczna 110kV „Koksownia Zdzieszowice"** in Opolskie, Opolskie. With an R_final of 0.507 (High band, 70th fleet percentile), its risk profile is dominated by the T (Transition) component at 698% of its weight ceiling, followed by V (Voltage) at 397%. Modifiers collectively shift R_base by 16.9%%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Poland's Energy Regulatory Office (URE) and network operator strategic planning requires.

CYBER HIGH EXPOSURE
0
0.0% of fleet

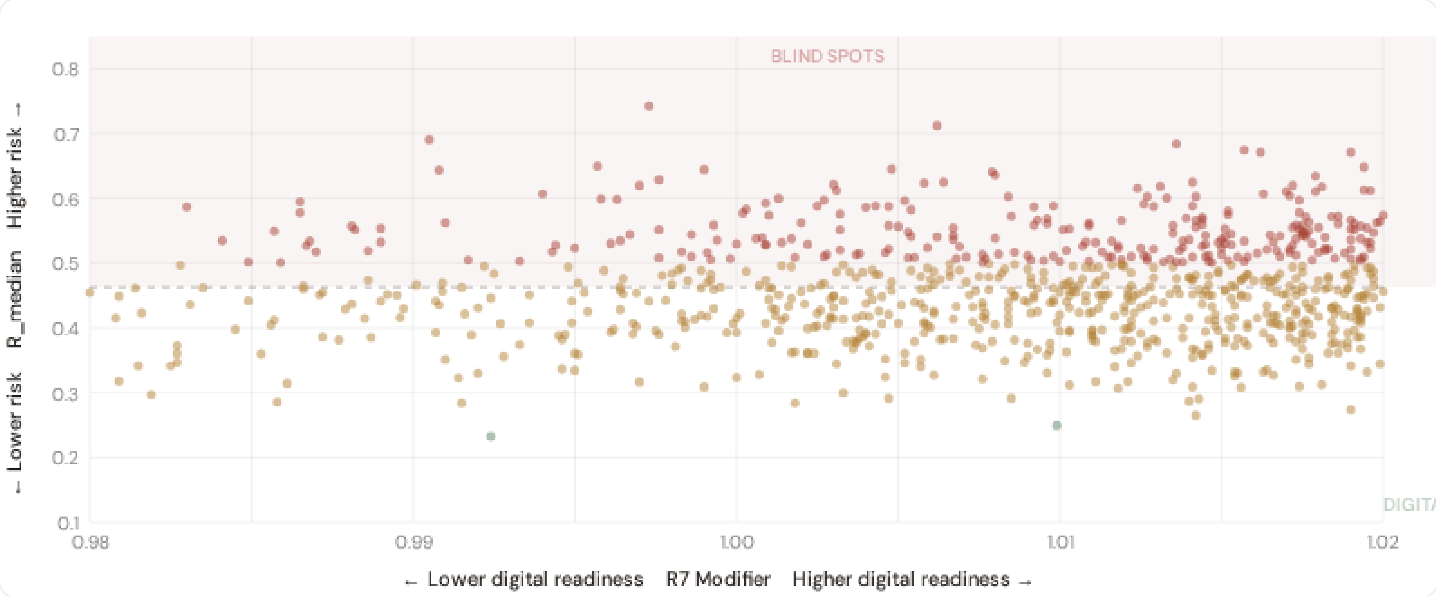
BLIND SPOTS
341
High/Critical + R7 < median

AVG R7 MODIFIER
1.0252
Range [0.99, 1.05]

HIGH-CONSEQUENCE SUBS
565
25.1% · R3 ≥ 1.08

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **341 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 1.0256). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Śląskie (77), Dolnośląskie (48), Kujawsko-Pomorskie (29)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Industrial Sector

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

🏭 Capital-Intensive / High-Consequence

Est. VoLL: zł60–120/kWh (PLN)

Capital nodes — highest economic loss per interruption (semiconductor fabs, data centers, refineries, hospitals)

565 substations (25.1%) High/Critical: **275** (48.7%) Avg R: **0.499** Avg E: **0.389** / 0.10

Low 0 Med 290 High 273 Crit 2

🔧 Industrial / Medium-Large Enterprise

Est. VoLL: zł30–60/kWh (PLN)

Industrial belt — significant continuity requirements (manufacturing, logistics hubs)

569 substations (25.3%) High/Critical: **211** (37.1%) Avg R: **0.475** Avg E: **0.394** / 0.10

Low 1 Med 357 High 210 Crit 1

🏪 Commercial / SME-Dense

Est. VoLL: zł12–30/kWh (PLN)

Service economy + retail — moderate individual impact (commercial districts, mixed-use)

562 substations (25.0%) High/Critical: **142** (25.3%) Avg R: **0.456** Avg E: **0.379** / 0.10

Low 0 Med 420 High 142 Crit 0

🌾 Light / Agricultural / Rural

Est. VoLL: zł4–12/kWh (PLN)

Sparse activity, agricultural land — lower VoLL (residential, agricultural)

552 substations (24.6%) High/Critical: **112** (20.3%) Avg R: **0.436** Avg E: **0.391** / 0.10

Low 1 Med 439 High 112 Crit 0

Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **275 substations** combine capital-intensive economic fabric (R3 ≥ 1.08) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Dolnośląskie (145), Śląskie (95), Małopolskie (67)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 6.1%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Voivodeship & District Digital Readiness Ranking

Voivodeships ranked by average R7 modifier — lower values indicate weaker digital infrastructure (broadband penetration, smart meter coverage). Cross-referenced with average R_median to identify regions combining digital exposure with high grid stress.

WORST DIGITAL READINESS — TOP 15 REGION REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	Warmińsko-Mazurskie	1.0186
2	Pomorskie ⚠ high R	1.0221
3	Kujawsko-Pomorskie ⚠ high R	1.0228
4	Dolnośląskie ⚠ high R	1.0243
5	Podkarpackie ⚠ high R	1.0250
6	Małopolskie ⚠ high R	1.0251
7	Wielkopolskie	1.0251
8	Opolskie ⚠ high R	1.0253
9	Łódzkie	1.0253
10	Mazowieckie	1.0254
11	Podlaskie	1.0256
12	Świętokrzyskie	1.0258
13	Śląskie ⚠ high R	1.0260
14	Zachodniopomorskie	1.0270
15	Lubuskie ⚠ high R	1.0271

region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Warmińsko-Mazurskie** has the lowest average R7 (1.0186), while **Lubelskie** leads at 1.0292 — a spread of 0.0106 across the R7 range. **6 region regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

> **Inaugural edition — Baseline Snapshot (June 2026).** This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0252, median = 1.0256; 0 substations classified HIGH cyber-exposure; 341 blind-spot substations (High/Critical risk + below-median R7); 1 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **— verified public data sources** feeding — variables across — components and — modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES	WEEKLY / MONTHLY	QUARTERLY / ANNUAL
—	0	0
— variables	High-frequency feeds	Regular refresh cycle

STATIC / DERIVED
0
Standards + scenarios

Data Source Registry — June 2026

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

C.2 Fleet Score Snapshot

FLEET SIZE	MEDIAN R	HIGH + CRITICAL
2,248	0.463	740
substations scored	P5=0.342 · P95=0.603	32.9% of fleet

HIGH CONFIDENCE

100%

2248 subs · CI/R < 0.50

BAND DISTRIBUTION

Low 2 (0.1%) | Medium 1506 (67.0%) | High 737 (32.8%) | Critical 3 (0.1%)

No primary data sources have been updated since the v4.0.2 launch baseline. All **2,248 substation scores remain unchanged** this edition. The fleet median R of 0.463 (mean 0.467) reflects a distribution skewed toward the lower bands, with 0.1% in Low and 67.0% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

0 changes tracked across P0, P1, and P2 releases

SECTION D — MONTHLY DEEP-DIVE

The Śląskie–Mazowieckie Industrial Corridor: Where Coal Transition Strains Grid Resilience

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Śląskie–Mazowieckie coal transition corridor · **002** Eastern Poland DER & EU funding mismatch · **003** Mining subsidence infrastructure stress · **004** Metropolitan-Rural energy poverty disparity · **005** Grid stranding risk (coal-dependent assets) · **006** DSO smart meter rollout vs grid stress · **007** Vistula flooding resilience · **008** T-component forward projections · **009** Voivodeship Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Why the Śląskie–Mazowieckie Corridor, Why Now

CORRIDOR SUBSTATIONS

160

10 EHV · 150 HV

HIGH BAND

85 + 2

54.4% of corridor

CORRIDOR MEDIAN R

0.510

Fleet median: 0.463

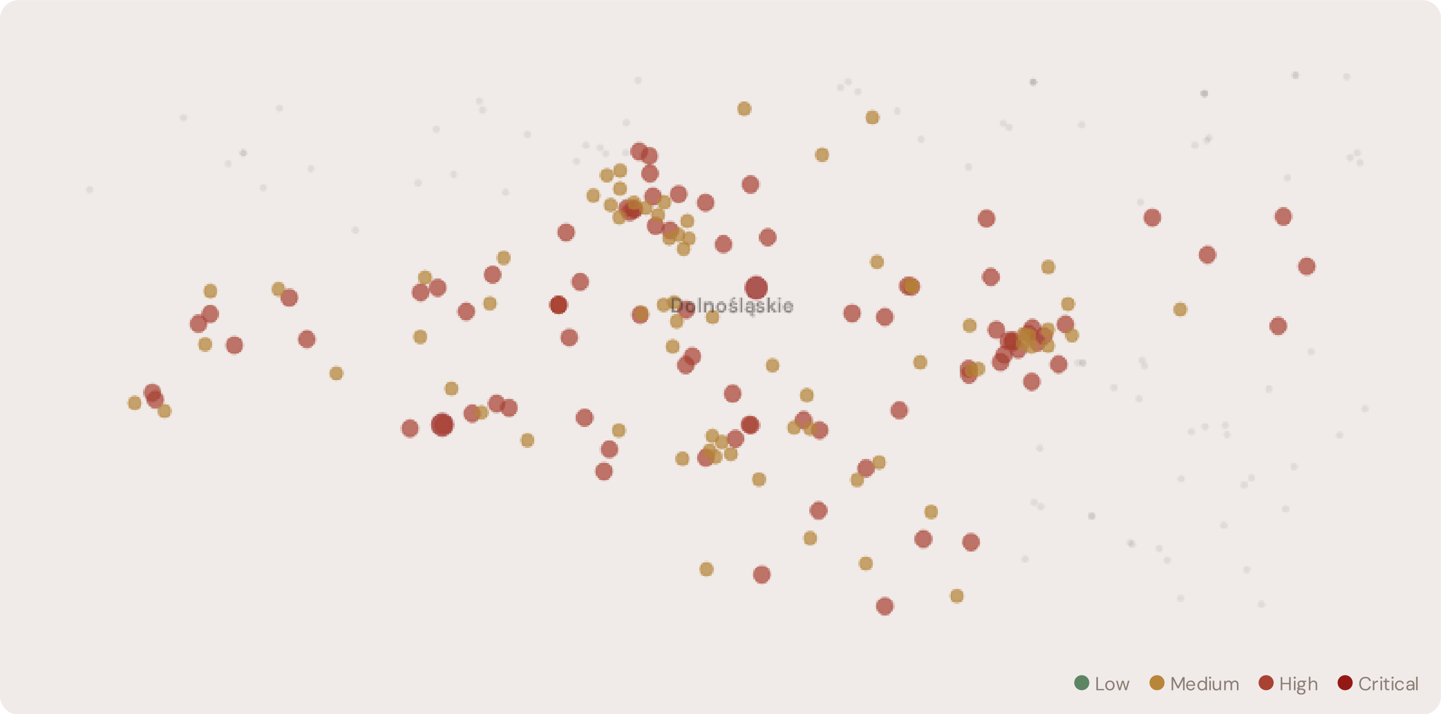
WORST VOIVODESHIP

Dolnośląskie

Avg R: 0.514 · 160 substations

The Dolnośląskie hosts **160 substations** across 1 region regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **85 substations (53.1%)** are classified High and **2** Critical, with only **0** in the Low band. 75% of corridor substations score above the national fleet median of 0.463. The corridor median R of 0.510 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring



SUBSTATION	DISTRICT	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Stacja elektroenergetyczna 110/20 kV "Piechowice"	Dolnośląskie	0.797	Critical	0.697	0.361	0.802	0.317	0.663	0.402	I
Stacja elektroenergetyczna 110kV Prochowice	Dolnośląskie	0.757	Critical	0.649	0.643	0.695	0.318	0.546	0.270	
Stacja elektroenergetyczna "FW Łukaszów" 110kV	Dolnośląskie	0.712	High	0.603	0.499	0.757	0.254	0.577	0.395	I
Stacja elektroenergetyczna 110/20kV "Bystrzyca"	Dolnośląskie	0.675	High	0.553	0.682	0.594	0.396	0.676	0.301	
Stacja elektroenergetyczna 110/20kV "Graby"	Dolnośląskie	0.644	High	0.346	0.442	0.683	0.543	0.559	0.322	
Stacja elektroenergetyczna "Bolesławiec Tysiąclecia" 110/20 kV	Dolnośląskie	0.625	High	0.643	0.485	0.489	0.355	0.399	0.336	

SUBSTATION	DISTRICT	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Stacja elektroenergetyczna 110kV „Victoria”	Dolnośląskie	0.625	High	0.503	0.412	0.622	0.427	0.412	0.190	
Stacja elektroenergetyczna 400/220/110kV „Mikułowa”	Dolnośląskie	0.624	High	0.563	0.451	0.603	0.559	0.388	0.235	
Stacja elektroenergetyczna 400/220/110kV „Polkowice”	Dolnośląskie	0.620	High	0.392	0.240	0.694	0.454	0.466	0.432	
Stacja elektroenergetyczna 110/20kV „Dzierżoniów”	Dolnośląskie	0.618	High	0.571	0.344	0.537	0.575	0.361	0.176	

… 150 additional substations — explore full corridor in Digital Twin →

D.3 Component Analysis

COMPONENT AVERAGES — DOLNOŚLĄSKIE VS FLEET

C — Continuity 0.369 vs 0.382 (-3%)

I — Infrastructure 0.468 vs 0.416 (+12%)

S — Saturation 0.453 vs 0.456 (-0%)

V — Voltage 0.429 vs 0.376 (+14%)

E — Economic 0.361 vs 0.388 (-7%)

T — Transition 0.312 vs 0.309 (+1%)

MODIFIER AVERAGES — DOLNOŚLĄSKIE VS FLEET

R3 Consequence	1.101	fleet 1.060 ↑
R4 Graph Crit.	1.008	fleet 1.010 →
R6a Restoration	0.996	fleet 0.996 →
R6b Seismic	1.099	fleet 1.064 ↑
R7 Digital Read.	1.024	fleet 1.025 ↑

The dominant risk driver across the Dolnośląskie corridor is **T (Transition)**, averaging 0.312 against a fleet average of 0.309 — occupying 624% of its weight ceiling (w = 0.05). The second contributor is **V (Voltage)** at 0.429 vs fleet 0.376 (429% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.101 (fleet: 1.060), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 1.099, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 Voivodeship Risk Profile

REGION RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Dolnośląskie 160 subs · avg R = 0.514 · H:85 C:2 M:73

The region regions-level breakdown reveals significant intra-regional variation. **Dolnośląskie** leads with a mean R of 0.514 across 160 substations, while **Dolnośląskie** scores 0.514 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Dolnośląskie is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications for Poland's Coal Transition

4 Dolnośląskie substations score R ≥ 0.650, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Dolnośląskie sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

SECTION E — RECURRING MODULE

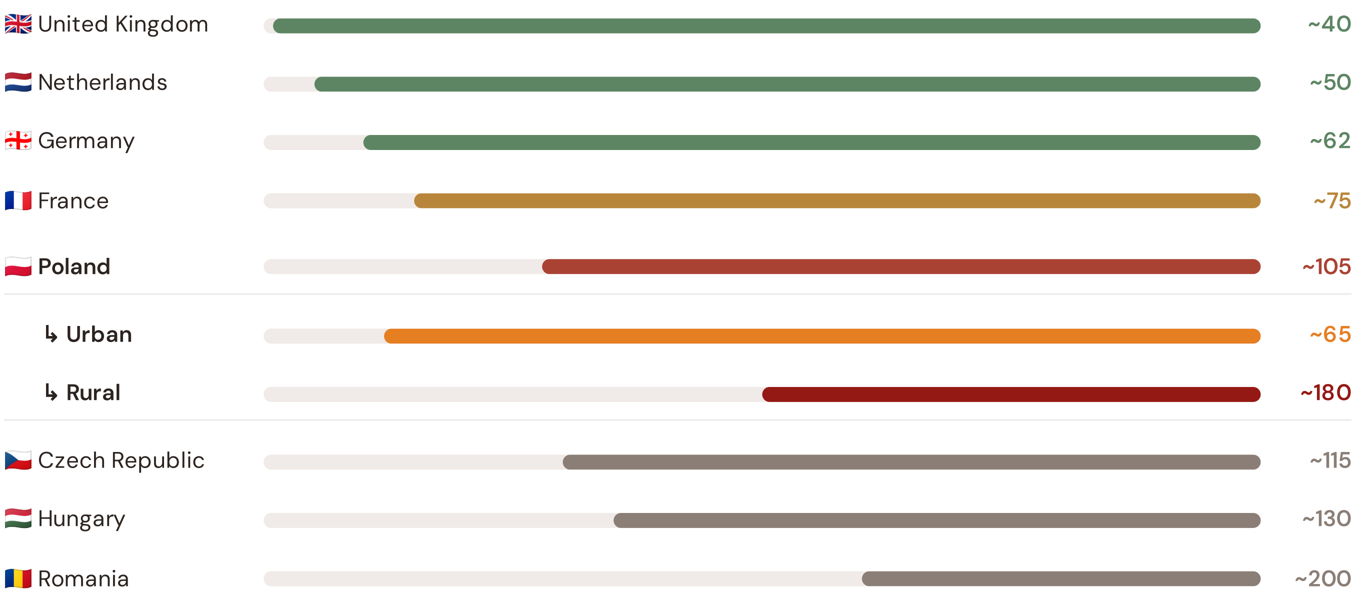
European Context Card

> **How this works:** Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because Poland's coal transition and grid resilience depend on understanding European performance standards. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Central Europe peers (003), CMIP6 climate hazard vs Nordic models (004), etc. The underlying data updates annually with URE and national regulator publications.

EUROPEAN CONTEXT CARD — SYSTEM AVERAGE INTERRUPTION DURATION INDEX Ed. 001 · April 2026

Unplanned SAIDI (min/customer/year) — Selected European Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



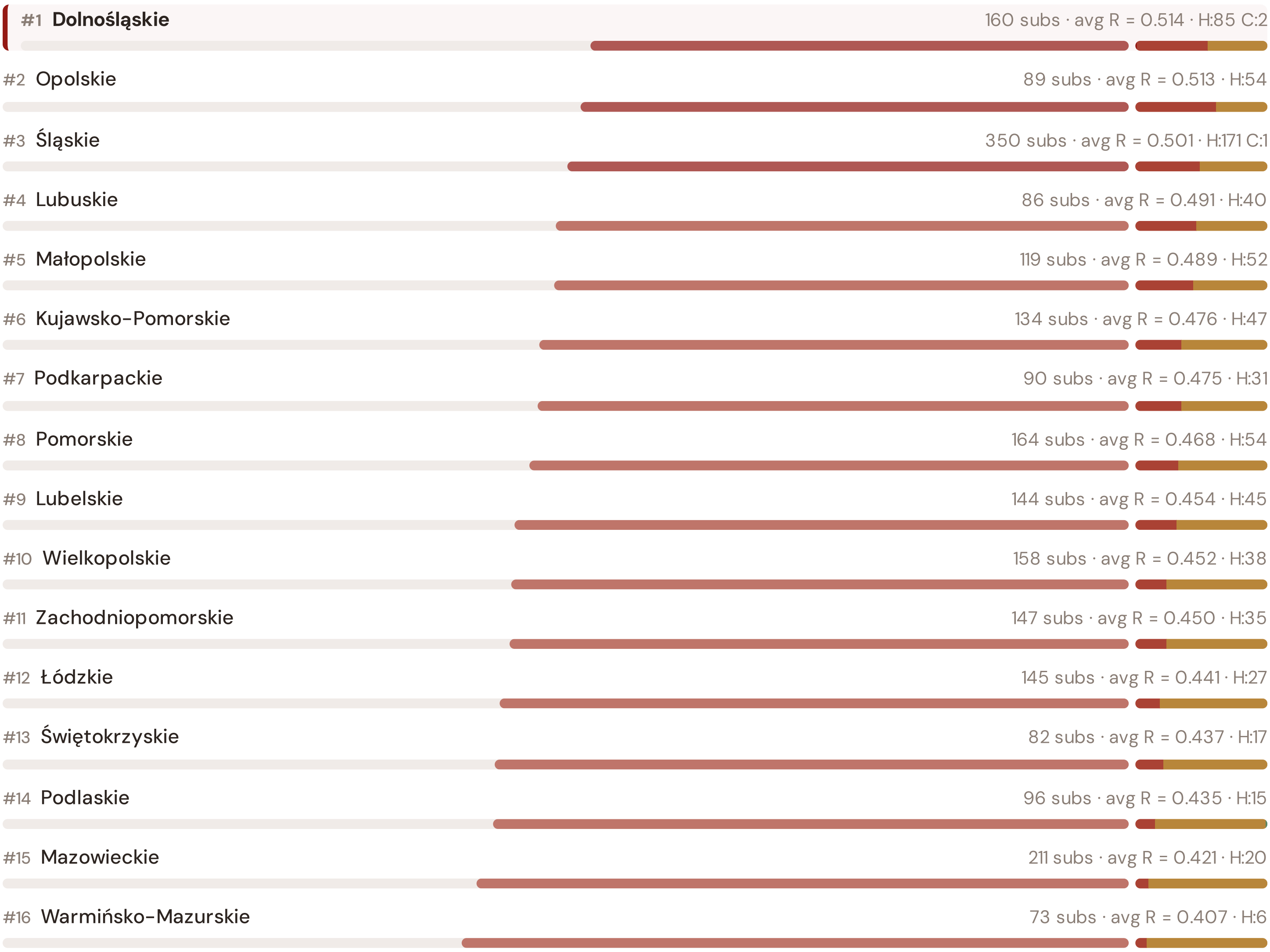
Source: CEER Benchmarking Report on Quality of Electricity Supply (2024); EURELECTRIC SAIDI reports; national regulator publications. Poland sub-categories from URE urban/rural classifications. · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually.

This month's key insight: The Dolnośląskie corridor deep-dive shows **137 substations** (of 160) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.369 is **1.0× the fleet average** of 0.382. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Voivodeship Risk Ranking — All 16 Polish Voivodeships

Where does the Śląskie–Mazowieckie corridor sit within the national landscape? Voivodeships sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



The Dolnośląskie ranks **#1 of 16 region regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed are Dolnośląskie, Opolskie, Śląskie, while the three most resilient are Warmińsko-Mazurskie, Mazowieckie, Podlaskie. 8 of 16 score above the fleet average of 0.467. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to renewable energy investment valuation. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a battery energy storage or renewable retrofit worth at this substation — to the DSO/PSE and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

Layer 1 — Data	Layer 2 — Valuation	Layer 3 — Neural Net	Layer 4 — Coverage	Layer 5 — Platform
SSI v4.0 component taxonomy, normalisation, MC engine	DSO/TSO + Community value stacks, TCO, real options	GBT+DNN ensemble, transfer learning, SHAP	Country prioritisation, DCI, registry, calibration	Products, API, revenue model, unit economics

LAYER 3 §0.5.6 · June 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history. Transparency converts model outputs into investment decisions.

KEY CONCEPTS

- → SHAP: game-theoretic feature attribution per prediction
- → Waterfall: fleet average → substation prediction
- → Direct traceability to SSI components and data sources
- → Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The training set comprises 2,248 substations with complete feature vectors. Average CI_width of 0.162 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 2 Low, 1506 Medium, 737 High, 3 Critical — ensures balanced training across all risk bands.

Coming next month: LAYER 3 Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 02

Mountain Region

Deep-dive into Region 10's grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Light-rural tier — limited substation density.

NEXT DATA REFRESH

Q3 2026 — 0 Quarterly Sources

none are due for refresh in Q3. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **TBD** (? variables →).

FLEET SPOTLIGHT

Opolskie: Highest Risk Concentration

Opolskie has the highest concentration of High/Critical substations: **54 of 89 (61%)** are in the upper risk bands — avg R = 0.513. This makes it the most acute risk corridor in the fleet.

Edition 02 will be published on the second Thursday of July 2026. Deep-dive region: **Region 10**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.